

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester of M.Sc. (Mathematics) Examination November 2018

MA711 Complex Analysis

Date: 26/11/2018, Monday

Time: 10:00 a.m. to 10:45 a.m. Maximum Marks: 20

Multiple Choice Questions

Important Instructions:

1. Tick the correct answer and it should be written in question paper itself.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Each question carries one mark.
4. All notations are standard.

Q - I Choose the correct answer from the given options for the followings: [20]

1. For $f(z) = u + iv$, the polar form of $f'(z) =$ _____.
(a) $u_r + iv_r$ (b) $e^{-i\theta}(u_r + iv_r)$ (c) $e^{-i\theta}(u_\theta + iv_\theta)$ (d) $r(u_r + iv_\theta)$
2. $\sin z$ is a periodic function with period _____.
(a) 2π (c) $k\pi/2, k \in \mathbb{Z}$
(b) $(2k+1)\pi/2, k \in \mathbb{Z}$ (d) $k\pi, k \in \mathbb{Z}$
3. If $\log z = \log r + i\theta, r > 0$ and $\frac{3\pi}{4} < \theta < \frac{11\pi}{4}$, then _____.
(a) $\log i^2 \neq 2 \log i$. (c) $\log z$ is not defined
(b) $\log i^2 = 2 \log i$. (d) None of these
4. $\text{Arg}(-\sqrt{2} - i) =$ _____.
(a) $\frac{5\pi}{6}$ (b) $\frac{\pi}{6}$ (c) $-\frac{5\pi}{6}$ (d) $-\frac{\pi}{6}$

5. The equation $z - \cos(\text{Log} i^i) = 0$ has _____.
 (a) not a unique solution (c) a root -1
 (b) a root 1 (d) a root 0
6. $z = -1 - i$ has an exponential form _____.
 (a) $\sqrt{2}(e^{-3\pi/4+2n\pi})$, $n \in \mathbb{Z}$ (c) $\sqrt{2}(e^{i(-\pi/4+2n\pi)})$, $n \in \mathbb{Z}$
 (b) $\sqrt{2}(e^{i(-3\pi/4+2n\pi)})$, $n \in \mathbb{Z}$ (d) None of these
7. The function $f(z) = |z|$ is _____.
 (a) entire (c) nowhere differentiable
 (b) analytic (d) None of these
8. For $z = x + iy$, $\arg(e^z) =$ _____.
 (a) $y + 2k\pi$, $k \in \mathbb{Z}$ (c) $x + 2k\pi$, $k \in \mathbb{Z}$
 (b) $\tan^{-1}(\frac{y}{x})$ (d) None of these
9. The function $f(z) = \text{Arg}(z)$ is _____.
 (a) multi-valued (c) not continuous
 (b) differentiable (d) None of these
10. If C denotes the positively oriented circle $|z - 2i| = 1$, then $\int_C (z - 2i)^2 dz =$ _____.
 (a) $2\pi i$ (b) 0 (c) 2π (d) $\frac{1}{2\pi i}$
11. $\int_{|z|=1} \frac{ze^{-z} dz}{z - 3} =$ _____.
 (a) 0 (b) 2 (c) 1 (d) None of these
12. The radius of convergence of the series $\sum_{n=0}^{\infty} (n + 6)^n z^n$ is _____.
 (a) ∞ (b) e (c) 0 (d) None of these
13. The series $\sum_{n=0}^{\infty} (\log n)^n 2^n$ is _____.
 (a) convergent (c) oscillating
 (b) not absolutely convergent (d) None of these
14. An isolated singular point of $\log z$ is _____.
 (a) 0 (c) i
 (b) -1 (d) None of these

15. For $|z| > 2$, the function $f(z) = \underline{\hspace{2cm}}$ has the series expansion $\frac{-1}{z} \left[1 + \frac{2}{z} + \frac{2}{z^2} + \frac{2}{z^3} + \cdots \right]$.

(a) $\frac{1}{2-z}$

(c) $\frac{-z}{2-z}$

(b) $\frac{1}{z-2}$

(d) $\frac{z}{2-z}$

16. The Taylor's series of $f(z) = \frac{1}{z}$ about the point $z = 3$ is $\underline{\hspace{2cm}}$.

(a) $\sum_{n=0}^{\infty} \frac{(-1)^n}{3^{n+1}} (z-3)^n, |z+3| < 3$

(c) $\sum_{n=0}^{\infty} \frac{(-1)^n}{3^{n+1}} (z-3)^n, |z| < 3$

(b) $\sum_{n=0}^{\infty} \frac{(z-3)^n}{3^{n+1}}, |z-3| < 3$

(d) $\sum_{n=0}^{\infty} \frac{(-1)^n}{3^{n+1}} (z-3)^n, |z-3| < 3$

17. The function $f(z) = z^3 e^{\frac{1}{z}}$ has $\underline{\hspace{2cm}}$.

(a) removable singularity

(c) essential singularity

(b) simple pole

(d) None of these

18. The function $f(z) = \frac{e^z}{(z-1)^3}$ has $\underline{\hspace{2cm}}$.

(a) a residue, $\frac{1}{2!}$

(c) a pole at $z = 1$ of order 4

(b) a pole at $z = 1$ of order 3

(d) a zero at $z = 0$

19. The map $w = \underline{\hspace{2cm}}$ is not a valid Möbius transformation.

(a) $\frac{3+4z}{10z+5}$

(c) $\frac{6+5z}{11z-12}$

(b) $\frac{1+2z}{4z+1}$

(d) None of these

20. The function $f(z) = \bar{z}$ is $\underline{\hspace{2cm}}$ mapping.

(a) an isogonal

(c) a non-isogonal

(b) a conformal

(d) None of these

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Date: 26/11/2018, Monday Time: 10:45 a.m. to 01:00 p.m. Maximum Marks: 50

Instructions:

1. **Section I and Section II must be written in separate answer books..**
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. All notations are standard.

SECTION I

Q - II (A) Attempt any TWO of the followings: [8]

- (1) The statement “If a function satisfies Cauchy-Riemann equations then it is a differentiable function.” is true or false? Justify.
- (2) Define harmonic function. Show that $u(x, y) = \sinh x \sin y$ is harmonic. Also find its harmonic conjugate $v(x, y)$ such that the function $f = u + iv$ is an analytic function.
- (3) State ML-inequality and hence show that if C is the positively oriented boundary of the triangle with vertices at $z = 0, 3i, -4$, then show that

$$\left| \int_C (e^z - \bar{z}) dz \right| \leq 60.$$

(B) Attempt any SIX of the followings: [12]

- (1) State and prove the triangle inequality for the complex numbers.
- (2) If $\overline{(e^{iz})} = e^{i\bar{z}}$ then prove that $z = n\pi, n \in \mathbb{Z}$.
- (3) Find all the solutions of the equation $z^4 = -4$.
- (4) Show that $\log i^2 = 2 \log i$, where $\log z = \ln |z| + i \arg z$ with $\frac{\pi}{4} < \arg z < \frac{9\pi}{4}$.
- (5) If the power series $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ converges at $z = z_1, z_1 \neq z_0$, then prove that it is absolutely convergent at each point z in the open disk $|z - z_0| < R_1$ where $R_1 = |z_1 - z_0|$.
- (6) State and prove Morera's theorem.
- (7) Find the residue of the function $\frac{e^z}{(z-1)^3}$.
- (8) Prove that the Möbius transformation is an onto function.
- (9) Define conformal mapping. Also give its example.

SECTION II

Q - III (A) Attempt any THREE of the followings:

[18]

- (1) Let the function $f(z) = u(x, y) + iv(x, y)$ be defined throughout some ϵ -neighborhood of a point $z_0 = x_0 + iy_0$. Prove that f is differentiable at z_0 if and only if the first order partial derivatives of functions u and v exist in this neighborhood, they are continuous at (x_0, y_0) and they satisfy Cauchy-Riemann equations at (x_0, y_0) .
- (2) State and prove Cauchy integral formula and hence evaluate $\int_C \frac{\cos z \, dz}{z(z^2 + 8)}$, where C denotes the positively oriented boundary of the square whose vertices are $(\pm 2, \pm 2)$.
- (3) If a function f is analytic and non-constant in a given domain D , then prove that $|f(z)|$ does not have maximum value in D .
- (4) State and prove the residue theorem and hence evaluate $\int_{|z|=3} \frac{z+1}{z^2-2z} dz$.
- (5) Prove that an angle of rotation is produced by the transformation $w = \frac{1}{z}$ at the point $z = 1$ is π .

(B) Attempt any TWO of the followings:

[12]

- (1) Using residue theory, evaluate the improper integral $\int_0^\infty \frac{dx}{x^4-1}$.
- (2) Obtain the Laurent series expansion of the function $f(z) = \frac{z+4}{(z+3)(z-1)^2}$, valid in (i) $0 < |z-1| < 4$ (ii) $|z-1| > 4$.
- (3) If $k < 0$, show that the image of the half plane $x < k$ under the transformation $w = \frac{1}{z}$ is an interior of the circle. What is the image when $k = 0$ under the same transformation?
